Basic Number Concepts: Place Value, Decimals & Fractions

Overview

This document converts the provided notes into a clean LaTeX source. It covers:

- Place value (whole numbers and decimals),
- Decimals and fractions (terminology and conversions),
- Comparing and ordering numbers (vocabulary and symbols),
- Prime and composite numbers (prime factorization and divisibility tests),
- Greatest common factor (GCF) and least common multiple (LCM) with the "ㄴ" method.

1 Place Value

1.1 Essential whole-number place values

The important whole-number place values to learn (from largest to smallest) are:

English	Korean	Value (power of 10)
billions	십억 (or 억, depending on grouping)	10^9
hundred millions	백억	10^8
ten millions	십억	10^7
millions	백만	10^6
hundred thousands	십만	10^5
ten thousands	만 (만 = 10^4 in Korean system)	10^4
thousands	천	10^3
hundreds	백	10^2
tens	십	10^1
ones	일	10^0

Note on Korean grouping: In Korean the grouping differs from English after the thousands place. Korean uses units of 만 (10^4), 억 (10^8), 조 (10^{12}), etc. The user notes: "한국어와 비슷하게 앞자리에 10배만 붙이면 됨: 일 십 백 천 만 십만 백만 천만 억 십억 백억" — this is a mnemonic showing successive $10\times$ steps and how the Korean names appear.

1.2 Decimal place values

Decimal places (to the right of the decimal point) are named by adding "-ths":

- tenths (할 in Korean) $= 10^{-1}$,
- hundredths (푼) $= 10^{-2}$,
- thousandths (리) $= 10^{-3}$,
- ten-thousandths $= 10^{-4}$, and so on.

The notes recommend memorizing tenths, hundredths, and thousandths especially.

2 Decimals and Fractions

2.1 Terminology

Numerator (분자): the top part of a fraction.

Denominator (분모): the bottom part of a fraction.

If remembering the words is difficult, the user suggests the mnemonic "AND" to recall the order numerator/denominator ("and" = n/d order).

2.2 Types of fractions

- **Improper fraction (가분수)**: numerator $\geq$ denominator (e.g. $7/4$).
- **Mixed number (대분수)**: an integer plus a proper fraction (e.g. $1\frac{3}{4}$).

The suggestion: understanding the property is usually enough to remember the terms.

2.3 Types of decimals

- **Repeating / recurring decimal (순환소수)**: digits repeat periodically, often notated with a bar (e.g. $1.\overline{3} = 1.333\ldots$).
- **Finite decimal (유한소수)**: a decimal with a finite number of nonzero digits after the point (e.g. 0.25).
- **Infinite decimal (무한소수)**: non-terminating and non-repeating decimals; these are generally not expressible as fractions.

2.3.1 Detecting repeating decimals in notation

A repeating decimal is often indicated by placing a dot or bar over the repeating digits. Examples:

- $1.\dot{3} = 1.333\ldots = \dfrac{4}{3}$.
- If the digits 3475 repeat in $1.34753475\ldots$, it may be written with dots or a vinculum over the repeating block.

2.3.2 Converting a repeating block to a fraction

If a repeating block has k digits and starts immediately after the decimal point, the repeating portion equals an integer over $10^k - 1$ (a string of k nines). For example, if $x = 0.\overline{3984}$ then

$$x = \frac{3984}{9999}.$$

If the repeating block starts after some non-repeating digits, handle by subtracting the non-repeating portion or using standard algebraic techniques (noted above as the usual method).

3 Comparing and Ordering Numbers

Common comparison symbols and English vocabulary:

$>$	greater than (more than, larger than, higher than, above, exceeds, over)
$\geq$	greater than or equal to (at least, minimum, no less than)
$<$	less than (fewer than, smaller than, lower than, under, below)
$\leq$	less than or equal to (at most, maximum, no more than)

The notes suggest focusing on the most test-relevant phrases (e.g. "at least/at most") and to read each problem carefully to interpret the intended meaning.

4 Prime and Composite Numbers

4.1 Prime factorization

Prime factorization is the process of breaking a positive integer into a product of prime numbers (its smallest multiplicative building blocks).

Basic procedure:

1. Check small prime divisors (2, 3, 5, 7, ...) using divisibility tests.
2. Divide repeatedly by primes until the remaining factor is 1.

4.2 Common divisibility tests (useful heuristics)

- **Divisible by 2:** last digit is even (0,2,4,6,8).
- **Divisible by 3:** sum of digits is divisible by 3.
- **Divisible by 4:** last two digits form a number divisible by 4 (or end with 00).
- **Divisible by 5:** last digit is 0 or 5.
- **Divisible by 7:** a quick test is: for a three-digit grouping abc, check whether $ab - 2c$ is divisible by 7 (this is optional / rarely needed).

4.3 Example: Factorize 60

$$60 \div 2 = 30, \quad 30 \div 2 = 15, \quad 15 \div 3 = 5, \quad 5 \div 5 = 1.$$

Hence the prime factorization is

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \cdot 3 \cdot 5.$$

5 Greatest Common Factor (GCF) & Least Common Multiple (LCM)

The notes describe the Korean schoolbook "ㄴ" method for finding GCF and LCM of two or more numbers. The idea is to successively divide the numbers by common divisors and record the divisors.

5.1 Procedure ("ㄴ" method) and example

Find the GCF and LCM of 12 and 18.

Start by arranging the numbers side-by-side and dividing by a common factor repeatedly (here 2 then 3):

$$\begin{array}{r|cc} 6 & 12 & 18 \\ \hline & 2 & 3 \end{array}$$

In this representation the last row (to the right/below the dividing line) shows the quotients after dividing by the common factors; when the remaining numbers are prime (or no larger common factor remains) the process ends.

GCF: the product of the common divisors used in the vertical division process. In this case the common divisors are 2 and 3, so

$$\mathrm{GCF}(12, 18) = 2 \times 3 = 6.$$

LCM: the product of the GCF and any remaining quotients (equivalently, multiply all numbers that appear on the right and bottom of the ㄴ chart). Using the example above,

$$\mathrm{LCM}(12, 18) = 6 \times 2 \times 3 = 36.$$

(Another standard method: compute via prime factorizations and take the highest power of each prime that appears.)

Closing Notes

This LaTeX source faithfully converts the supplied learning notes into structured sections and examples. If the user prefers a different document class (`article` vs `report`), additional Korean font packages (for guaranteed correct Korean rendering), or a compiled PDF, indicate the preference and a downloadable file can be prepared.